
Decoding Keystrokes from Electromyography Signals: Architecture Selection, Ablation Studies, and Learned Front-End Design

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Abstract

We study the problem of decoding QWERTY keystrokes from surface electromyography (sEMG) signals for a single subject using the emg2qwerty dataset (2). Starting from the provided TDS CNN baseline (test CER 22.17%), we systematically evaluate encoder architectures, including vanilla RNN, BiLSTM, BiGRU, Transformer (3), Conformer (4), several hybrid designs, and find that compact gated recurrent models are the best fit for this low-data, single-user setting. We then conduct ablation studies on temporal resolution, context windowing, electrode channel count, training data amount, regularization, and data augmentation. Key findings include: (i) lower spectrogram frame rates (~ 41.7 Hz) outperform the baseline 125 Hz, (ii) stronger dropout with reduced SpecAugment yields the best regularization trade-off, and (iii) the data setting, not model capacity, is the primary performance bottleneck. Building on the temporal-resolution insight, we design a Raw-CNN+BiGRU encoder that learns temporal downsampling directly from raw EMG. Combined with CTC beam-search decoding and a character-level language model, our best system achieves a test CER of **6.81%** (spectrogram BiGRU) and **6.94%** (Raw-CNN+BiGRU), reducing the baseline error by over 69%.

Code Repository: https://github.com/mingwen1022/emg2qwerty_247a

1 Introduction

The emg2qwerty dataset pairs surface electromyography (sEMG) signals from both wrists with ground-truth keystrokes, enabling the development of typing decoders based on muscle activity. In this project, we focus on a single subject (#89335547) with 16 training sessions, a condition where data is limited and model efficiency is critical.

The baseline TDS-CNN encoder uses fixed-width convolutional kernels whose receptive field spans only ~ 62 ms, limiting its ability to capture the variable-timescale dependencies inherent in keystroke-related muscle activations. This raises a natural question: *can sequential models that see the full temporal context, such as recurrent or attention-based architectures, do better, and if so, which design choices matter most in a low-data condition?*

We structure our investigation around three phases. First, we compare a broad set of encoder architectures under controlled conditions to identify the best model family (§3.1). The central finding is that compact gated recurrent models (BiLSTM, BiGRU) consistently outperform both the CNN baseline and larger attention-heavy alternatives, suggesting that the bottleneck is data efficiency

rather than model capacity. Second, having selected recurrent encoders as the backbone, we conduct systematic ablation studies on temporal resolution, context design, regularization, data augmentation, channel count, and training data amount (§3.2). These ablations reveal that lower temporal resolution and stronger dropout, combined with *removal* of SpecAugment, yield the largest gains. Third, we pursue two complementary improvements: CTC beam-search decoding with a character language model, and a learned-downsampling Raw-CNN+BiGRU front end that replaces the fixed spectrogram (§3.3). Both provide substantial further reductions in CER, and the Raw-CNN design is directly motivated by the temporal-resolution insight from the ablation phase.

2 Methods

2.1 Experimental Setup

All experiments use the single-user split for subject #89335547 (16 training sessions, 1 validation, 1 test). We report Character Error Rate (CER) on both validation and test sets. Models are trained with CTC loss and decoded with greedy CTC decoding unless otherwise noted.

An important protocol detail: training and validation use windowed segments with configurable padding, while test evaluation feeds each full session without windowing. This asymmetry affects context-design experiments and is discussed in §3.2.2.

We adopt a two-stage protocol: **40-epoch screening** runs to efficiently explore the design space, followed by **150-epoch confirmation** runs for promising candidates. All screening-vs-confirmation distinctions are reported explicitly.

2.2 Pipeline Architecture

All spectrogram-based models share a common pipeline, differing only in the encoder:

SpectrogramNorm → MultiBandRotationInvariantMLP → Flatten
→ [Encoder] → Linear → LogSoftmax → CTCLoss

This controlled setup ensures that performance differences reflect encoder design rather than other pipeline components. The MLP, classifier head, and CTC decoder are held constant across all experiments. The Raw-CNN+BiGRU (§3.3.2) is the sole exception: it replaces the entire spectrogram front end with learned 1D convolutions applied to raw EMG.

2.3 Architecture Candidates

We evaluate the following encoder families:

- **CNN**: TDS-CNN baseline (4 blocks, kernel=32; 5.3M params).
- **Vanilla RNN**: BiRNN with tanh activation (5.0–9.5M params). Included as a negative control to isolate the effect of gating.
- **Gated recurrent**: BiLSTM and BiGRU with varying hidden sizes (384, 512) and depths (1–3 layers).
- **Hybrid**: TDS+BiLSTM (warm-started), Conv+BiLSTM+Conv, Conv+BiGRU+Conv.
- **Attention-based**: BiLSTM+Transformer (2 Transformer layers on top of BiLSTM), Conformer (2 layers, kernel=31), and standalone Transformer (d=768, 4 layers, 8 heads; 32.7M params).

2.4 Ablation Design

After selecting recurrent encoders as the best family, we conduct ablations on: (1) temporal resolution (hop length 8–64), (2) context window size and padding, (3) regularization (dropout 0.0–0.7, SpecAugment strength, optimizer choice), (4) data augmentation (Gaussian noise, amplitude scaling, band rotation range, temporal jitter), (5) electrode channel count (1–16 per band), and (6) training data amount (2–16 sessions). For channel and data-amount ablations, we additionally report Transformer results to enable cross-architecture comparison.

Table 1: 40-epoch Architecture comparison
All models use baseline preprocessing.

Architecture	Params	Val CER
<i>CNN baseline</i>		
TDS-CNN (k=32)	5.3M	22.55
<i>Recurrent (ungated)</i>		
BiRNN (h=512, l=3)	5.0M	51.66
CNN+BiRNN+2FC	9.5M	32.63
<i>Recurrent (gated)</i>		
BiLSTM (h=384, l=2)	8.2M	19.87
BiLSTM (h=512, l=3)	19.1M	17.88
BiGRU (h=384, l=2)	6.4M	20.56
<i>Hybrid</i>		
TDS+BiLSTM	13.0M	17.17
Conv+BiLSTM+Conv	9.4M	17.68
Conv+BiGRU+Conv	7.6M	27.49
<i>Attention-based</i>		
BiLSTM+Transf. (l=2)	22.3M	17.32
Conformer (l=2, k=31)	15.1M	30.77
Transformer (d=768, l=4)	32.7M	19.87

Table 2: 150-epoch confirmation runs
for top candidates. Baseline preprocessing.

Architecture	Params	Val CER	Test CER
TDS-CNN (k=32)	5.3M	18.94	22.17
BiRNN (h=512, l=5)	8.1M	45.70	37.00
CNN+BiRNN+2FC	9.5M	31.52	25.20
BiLSTM (h=384, l=2)	8.2M	14.55	15.76
BiLSTM (h=512, l=3)	19.1M	15.91	22.80
BiGRU (h=384, l=2)	6.4M	16.26	53.90
TDS+BiLSTM	13.0M	14.89	15.93
Conv+BiLSTM+Conv	9.4M	14.60	15.99
Conv+BiGRU+Conv	7.6M	14.84	16.92
BiLSTM+Transf. (l=2)	22.3M	14.67	17.25

3 Results

3.1 Results: Architecture Comparison (Phase 1)

We evaluate all encoder candidates under identical conditions of baseline preprocessing (hop=16, 125 Hz, win=8000), 40-epoch screening runs, and greedy CTC decoding. Table 1 reports validation CER for all architectures, we use validation performance to narrow down candidates, reserving test evaluation for 150-epoch confirmation runs (Table 2).

Gating is essential, not just recurrence. Ungated BiRNN performs poorly (val 51.66), and even with a CNN front end (32.63) it falls far behind gated alternatives.

Model size is not the bottleneck. At 40 epochs, the best screening val CER among gated models belongs to BiLSTM (h=512, 17.88) and TDS+BiLSTM (17.17). However, compact models such as BiLSTM (h=384, val 19.87) and BiGRU (h=384, val 20.56) achieve comparable performance with far fewer parameters (6–8M vs. 13–19M). Among attention-based models, the Transformer (32.7M params) reaches val 19.87, matching BiLSTM (h=384) with 4× the parameters, while the Conformer fails entirely (val 30.77).

Compact models generalize best under longer training. The 150-epoch confirmation runs (Table 2) reveal a notable ranking shift. BiLSTM (h=384, 8.2M) achieves test CER 15.76, while the larger BiLSTM (h=512, 19.1M) overfits (test 22.80), worse than the TDS-CNN baseline (22.17), despite a strong screening val of 17.88. Similarly, BiLSTM+Transformer (22.3M) converges to val 14.67 but generalizes poorly (test 17.25). These results confirm that in this data-limited condition, compact gated recurrent models generalize best, and screening val alone is not a reliable predictor of final test performance.

BiGRU (h=384, 6.4M) presents a striking case, it achieves a reasonable screening val of 20.56 but exhibits severe test instability under baseline preprocessing (test CER 53.90 at 150 epochs), suggesting high sensitivity to temporal resolution and context design. Resolving this instability is a key motivation for the ablation studies in §3.2, where optimized preprocessing transforms BiGRU into the best-performing encoder (test CER 6.81 with beam search). §3.3.1

Based on these findings, we select BiLSTM and BiGRU as the backbone for all subsequent ablation studies.

3.2 Results: Ablation Studies on Recurrent Encoders (Phase 2)

3.2.1 Temporal Resolution (Sampling Rate)

Table 3 summarizes the temporal resolution ablation. The baseline spectrogram uses hop=16 (125 Hz effective frame rate). We sweep hop lengths from 8 to 64 on BiLSTM. Counterintuitively, *lower* temporal resolution consistently outperforms higher rates. The best screening result is at hop=48 (~41.7 Hz), confirmed at 150 epochs (test CER 14.52 vs. baseline 20.08). The same trend holds for the Transformer: hop=32 (62.5 Hz) achieves 17.48 test CER vs. 20.92 at 125 Hz.

We hypothesize this reflects both sufficient temporal resolution for 50–200 ms keystroke activations and improved gradient flow from shorter sequences (discussed further in §4.2).

Table 3: Temporal resolution ablation.
BiLSTM (h=384, l=2, win=8000), 40-epoch screening except where noted.
Transformer results included for cross-architecture comparison.

Effective Rate	Architecture	Val CER	Test CER
250 Hz (hop=8)	BiLSTM	26.47	25.14
125 Hz (hop=16)	BiLSTM	19.87	20.08
83 Hz (hop=24)	BiLSTM	18.76	18.50
62.5 Hz (hop=32)	BiLSTM	17.50	16.99
41.7 Hz (hop=48)	BiLSTM	17.01	17.59
41.7 Hz (hop=48)	BiLSTM (150-ep)	13.98	14.52
31.25 Hz (hop=64)	BiLSTM	17.68	17.53
125 Hz (hop=16)	Transformer	19.87	20.94
62.5 Hz (hop=32)	Transformer	15.86	17.48

3.2.2 Context Design: Window Size and Padding

Table 4 summarizes the context design ablation. We ablate window length and contextual padding on BiLSTM with hop=48. Larger windows (16000 > 8000 > 4000) consistently help, as the BiLSTM benefits from more keystroke context per forward pass. However, *smaller* padding ([900,100]) outperforms the baseline [1800,200], and excessive padding [3600,400] degrades performance. The best combination is win=16000, pad=[900,100].

Table 4: Context design ablation.
BiLSTM (h=384, l=2, hop=48), 40-epoch screening except where noted.

Window / Padding	Notes	Val CER	Test CER
win=4000, pad=[1800,200]	short window	17.61	16.08
win=8000, pad=[1800,200]	baseline	17.01	17.59
win=16000, pad=[1800,200]	large window	15.86	16.19
win=8000, pad=[900,100]	smaller pad	15.77	16.06
win=8000, pad=[3600,400]	excess pad	17.39	17.22
win=16000, pad=[900,100]	best combo	15.13	16.27
win=16000, pad=[900,100]	150-ep confirm	13.65	14.93

3.2.3 Regularization: Dropout and SpecAugment

Table 5 summarizes the regularization ablation. We first vary dropout (0.0–0.7) on BiGRU with the best context configuration (hop=48, win=16000, pad=[900,100]) at 40 epochs, then confirm the best candidate at 150 epochs.

Dropout=0.5 is optimal at screening (val 14.33), with performance degrading at both lower and higher values. dropout=0.7 hurts significantly (val 14.98), indicating a non-monotonic relationship where excessive dropout destroys useful signal. The 150-epoch confirmation validates this, test CER drops from 14.11 (dropout=0.1) to 13.59 (dropout=0.5).

Table 5: Regularization ablation under greedy decoding.
BiGRU (hop=48, win=16000, pad=[900,100]). SpecAugment uses the baseline config unless noted.

Configuration	Horizon	Val CER	Test CER
dropout=0.0, SpecAugment	40-ep screen	14.93	-
dropout=0.1, SpecAugment	40-ep screen	15.37	-
dropout=0.2, SpecAugment	40-ep screen	14.53	-
dropout=0.3, SpecAugment	40-ep screen	14.51	-
dropout=0.5, SpecAugment	40-ep screen	14.33	-
dropout=0.6, SpecAugment	40-ep screen	14.38	-
dropout=0.7, SpecAugment	40-ep screen	14.98	-
dropout=0.1, SpecAugment	150-ep confirm	13.49	14.11
dropout=0.5, SpecAugment	150-ep confirm	13.03	13.59
dropout=0.5, no SpecAugment	40-ep screen	13.36	14.20
dropout=0.5, no SpecAugment	150-ep confirm	11.87	11.89

A critical finding follows from the SpecAugment ablation. We varied SpecAugment strength from the baseline (time_mask=25, freq_mask=4) down to very weak settings and complete removal. Removing SpecAugment entirely yields the largest single-factor improvement in the project, greedy test CER drops from 13.59 to 11.89 (−12.5% relative). We discuss why these regularizers interact negatively in §4.3.

3.2.4 Channel Count and Training Data Amount

Table 6 and 7 summarizes the channel count and training data ablations. These ablations address two suggested project directions and test the *data-bottleneck hypothesis* that emerged from the architecture experiments.

Channel count (Table 6). Performance degrades monotonically with fewer channels across both architectures. From 16 to 12 channels, BiLSTM test CER increases modestly (20.08 → 22.63), but below 8 channels the degradation accelerates sharply, CER roughly doubles with each subsequent halving (26.35 at 8ch, 37.97 at 4ch). This indicates that while a small degree of spatial redundancy exists among neighboring electrodes, each channel beyond ~8 still contributes unique muscle activation information. The Transformer degrades more steeply than BiLSTM (74.53 vs. 37.97 at 4 channels), fewer channels reduce the effective training signal, disproportionately hurting models that already struggle in this low-data setting.

Training data amount (Table 7). There is a sharp threshold effect. BiLSTM collapses completely below 8 sessions (CER ~100), while the Transformer degrades more gradually but is worse at all data levels. At 8 sessions, BiLSTM achieves 33.46 test CER, far from the 16-session result of 20.08, confirming that this setting is fundamentally data-limited.

Table 6: Channel count ablation.
40-epoch screening, 16 sessions, 125 Hz.

Ch/band	BiLSTM (8.2M)		Transformer (32.7M)	
	Val	Test	Val	Test
16	19.87	20.08	19.87	20.94
12	22.15	22.63	26.14	26.94
8	25.88	26.35	29.20	30.14
4	36.53	37.97	73.75	74.53

Table 7: Training data amount ablation.
40-epoch screening, 16 channels, 125 Hz.

Sessions	BiLSTM (8.2M)		Transformer (32.7M)	
	Val	Test	Val	Test
16	19.87	20.08	19.87	20.94
8	36.97	33.46	24.88	27.44
4	~100	~100	42.27	41.60
2	~100	~100	68.87	70.44

3.2.5 Data Augmentation

Beyond the regularization study, we tested additional augmentation strategies: Gaussian noise (std=0.1, post-spectrogram), amplitude scaling ($\times 0.7$ –1.3, pre-spectrogram), wider band rotation offsets, and varied temporal jitter. None improved over the baseline augmentation stack at 40 epochs. The baseline pipeline already contains three augmentation stages (band rotation, temporal jitter,

SpecAugment); adding further regularization slows convergence without improving generalization in this small-data condition.

3.2.6 Optimizer

We evaluated Adam ($\text{lr} \in \{3\text{e-}4, 5\text{e-}4, 1\text{e-}3\}$), AdamW ($\text{wd}=0.01$), and cosine-annealing schedules on the best BiGRU configuration. AdamW shows the best 40-epoch screening result (val 14.36) but fails to generalize at 150 epochs (test 15.32 vs. Adam’s 14.11). Cosine annealing causes test CER to explode at 40 epochs (~ 29), likely due to premature learning-rate decay. Adam with $\text{lr}=1\text{e-}3$ remains the best optimizer for this task.

3.3 Results: Decoder and Learned Front-End (Phase 3)

3.3.1 CTC Beam-Search Decoding

Table 8 summarizes the CTC Beam-Search ablations. We apply beam search with a character-level 6-gram LM to the best spectrogram BiGRU checkpoint ($\text{dropout}=0.1$, SpecAugment baseline). Beam search at $\text{beam}=50$ reduces test CER from 14.11 to 8.69, a 38% relative improvement without any encoder changes. Validation CER plateaus at $\text{beam}=75$ (8.40) with diminishing returns on test. We select $\text{beam}=50$ as the speed–accuracy trade-off.

When applied to the best regularized checkpoint ($\text{dropout}=0.5$, no SpecAugment), beam search achieves test CER 6.81, a further -16.6% relative improvement over the SpecAugment-enabled checkpoint (8.17), confirming that the regularization gains from §3.2.3 are additive to decoder improvements. This is the best overall result among spectrogram-based models.

Table 8: CTC beam-search decoding comparison ($\text{beam}=50$, 6-gram character LM).

Model / Configuration	Checkpoint	Val CER	Test CER
BiGRU, $\text{dropout}=0.5$, SpecAugment baseline	150-ep confirm	8.82	8.17
BiGRU, $\text{dropout}=0.5$, no SpecAugment	150-ep confirm	7.33	6.81
Raw-CNN+BiGRU (§3.3.2), $\text{gru_dropout}=0.0$	100-ep confirm	6.63	7.24
Raw-CNN+BiGRU, $\text{gru_dropout}=0.5$	120-ep confirm	6.72	6.94

3.3.2 Learned-Downsampling Raw-CNN+BiGRU

Table 9 summarizes the Learned Downsampling Raw-CNN+BiGRU ablation results. The temporal-resolution ablation (§3.2.1) showed that $\text{hop}=48$ (~ 41.7 Hz) is optimal. This motivates a natural question: instead of using a fixed spectrogram, can the model *learn* its own temporal downsampling from raw EMG?

The Raw-CNN+BiGRU replaces the spectrogram front end with three 1D convolutional layers (kernels [9,9,9], strides [3,4,4], paddings [4,4,4]) followed by a 2-layer BiGRU ($h=384$). The total stride is $3 \times 4 \times 4 = 48$, deliberately mirroring the best hop length. Channel counts increase as temporal resolution decreases ([96, 192, 256]), following the standard principle of trading spatial for channel capacity.

This model uses only ToTensor, no spectrogram, no SpecAugment, no band rotation, yet achieves greedy test CER of **11.48** (120 epochs, $\text{dropout}=0.5$), already outperforming the best spectrogram BiGRU under greedy decoding (11.89). With beam search (Table 8), it reaches test CER **6.94**, comparable to the spectrogram path (6.81).

Table 9: Raw-CNN+BiGRU results under greedy decoding.
Channels [96,192,256], win=20000, stride=12000, pad=[900,200].

Configuration	Horizon	Val CER	Test CER
$\text{gru_dropout}=0.0$	40-ep screen	10.64	12.60
$\text{gru_dropout}=0.5$	40-ep screen	10.31	12.45
$\text{gru_dropout}=0.0$	100-ep confirm	10.01	12.62
$\text{gru_dropout}=0.5$	120-ep confirm	9.93	11.48

4 Discussion

4.1 The Data-Bottleneck Hypothesis

The most consistent finding across all experiments is that this single-user setting is *data-limited*, not *capacity-limited*. Evidence converges from multiple directions: (i) scaling BiLSTM from 8.2M to 19.1M parameters worsens test CER (15.76 vs. 22.80); (ii) adding attention layers (Transformer, Conformer) adds parameters without improving generalization; (iii) the 6.4M BiGRU, once preprocessing is optimized, outperforms all larger models; (iv) reducing training sessions causes catastrophic failure below 8; and (v) stronger regularization (dropout=0.5) helps more than any architectural change.

This has a practical implication: for single-user EMG decoding with limited sessions, engineering effort is better spent on regularization and preprocessing than on architectural complexity.

4.2 Temporal Resolution: Why Less Is More

The consistent benefit of lower frame rates (~ 41.7 Hz vs. 125 Hz) can be understood through the timescales of the task. Keystroke-related muscle activations span 50–200 ms; at 41.7 Hz, each frame covers ~ 24 ms, providing more than sufficient temporal resolution. Higher rates produce longer sequences that (i) exacerbate vanishing gradients in recurrent encoders, (ii) increase frame-level noise, and (iii) add computational cost without informational benefit.

This insight directly motivated the Raw-CNN+BiGRU design. Rather than applying a fixed spectrogram transform, the model learns task-specific temporal compression with total stride 48, matching the empirically optimal hop length, while simultaneously learning spectral features from raw EMG. The result is not simply “a different model”; it is the logical endpoint of the ablation findings.

4.3 Competing Regularizers

The interaction between dropout and SpecAugment is the most practically useful finding from the ablation studies. At the default dropout=0.1, SpecAugment provides moderate regularization. But once dropout is increased to 0.5, the two compete: SpecAugment adds masking noise that the already-heavily-regularized network cannot learn through. Removing SpecAugment entirely yields the single largest improvement (-12.5% relative greedy CER). This suggests that in small-data conditions, regularization strategies should be tuned *jointly*, not stacked independently.

4.4 Decoder Gains Are Complementary

Beam search with a character LM provides a $\sim 38\%$ relative improvement on top of any encoder. Importantly, this gain is *additive* to encoder improvements: the best decoded spectrogram BiGRU (6.81) benefits from both a better encoder (dropout=0.5, no SpecAugment) and the language model. This decomposition, encoder improvements reduce acoustic confusion, while the LM corrects remaining errors using character-sequence priors, is well-established in speech recognition and transfers cleanly to EMG decoding.

4.5 Limitations

All results are from a single-user setting, the architecture ranking may not generalize across users. The Raw-CNN width search was intentionally narrow (two configurations) due to compute constraints. Heavily overlapped training windows can produce overly optimistic validation CER that does not transfer to full-session test evaluation (e.g., BiGRU with $\sim 83\%$ overlap: val 14.02, test 49.84), so care must be taken to match training and test windowing protocols. Best validation and test results sometimes come from different checkpoints, reflecting the small-data condition. Finally, beam search is applied offline; real-time deployment would require additional engineering.

5 References

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